Heat Conduction in a Thin Rod: Steady State and Stability

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Abstract

We look at how heat spreads along a thin rod whose ends are held at fixed temperatures, and at how a simple numerical scheme reproduces it.

1 Introduction

A thin rod conducts heat along its length. By Fourier's law the heat flux is proportional to $-\partial u/\partial x$, which leads to the heat equation of Section 2. The classical treatment is in [1].

2 Model

The temperature $u(x,t)$ obeys the heat equation

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}, \tag{1}$$

where $\alpha > 0$ is the thermal diffusivity. The ends are held at fixed temperatures:

$$u(0,t) = T_0, \tag{2}$$

$$u(L,t) = T_1. \tag{3}$$

As $t \to \infty$ the temperature settles to the steady state

$$u_s(x) = T_0 + (T_1 - T_0)\,\frac{x}{L}. \tag{4}$$

Theorem 1 (Maximum principle)**.** *The largest and smallest temperatures are reached at $t = 0$ or at the ends of the rod.*

Proof. At an interior maximum $u_{xx} \le 0$, so by (1) u cannot grow there. □

3 Results

We use the explicit scheme with $r = \alpha\,\Delta t/\Delta x^2 = 0.4$, which is stable because $r \le 1/2$ [2]. Table 1 lists the largest error at $t = 0.1$, and Figure 1 shows how the temperature approaches the steady state (4).

4 Discussion

The explicit scheme is simple, but its time step must shrink like Δx^2, which makes fine grids expensive. Implicit schemes such as Crank–Nicolson [3] avoid this limit:[1]

- they are stable for every $r > 0$, and
- they are second-order accurate in both Δt and Δx.

[1] At the cost of solving a tridiagonal system at every step.

Table 1: Largest error at $t = 0.1$, in units of 10^{-2}.

Δx	Error
0.1	12.88
0.05	3.19
0.025	0.79
0.0125	0.20

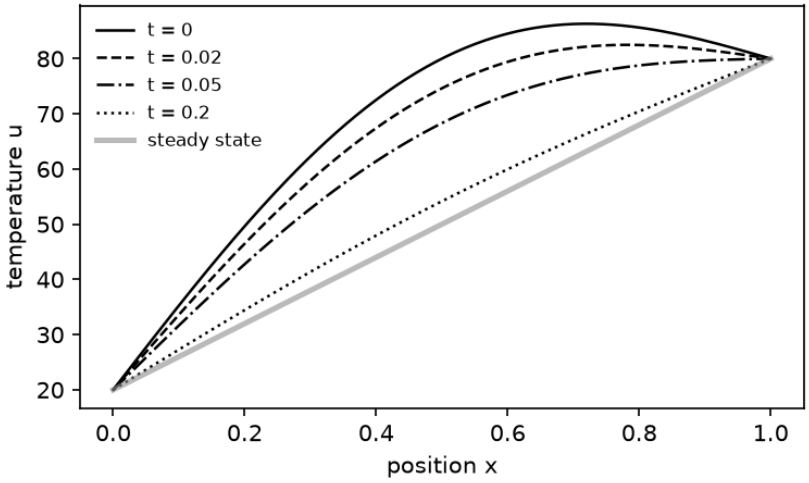


Figure 1: Temperature along the rod at $t = 0$, 0.02, 0.05 and 0.2, with the steady state.

References

[1] H. S. Carslaw and J. C. Jaeger, *Conduction of Heat in Solids*, Oxford University Press, 1959.

[2] G. D. Smith, *Numerical Solution of Partial Differential Equations: Finite Difference Methods*, Oxford University Press, 1985.

[3] J. Crank, *The Mathematics of Diffusion*, Oxford University Press, 1975.